

MBBS | COLLEGE LEVEL · BIOMEDICAL SCIENCES

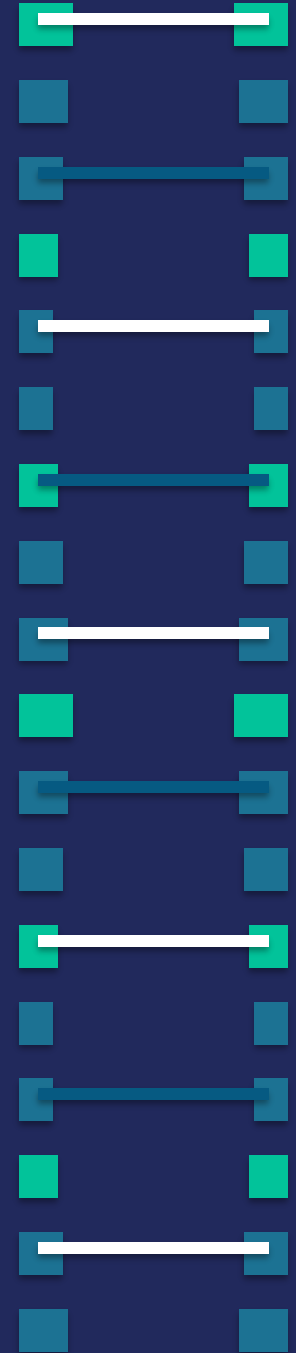
Gene Therapy & Genetic Engineering

A Comprehensive Overview for Medical & Life Science Students

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Introduction: Gene Therapy & Genetic Engineering

Gene Therapy

Introduction of therapeutic genetic material into cells to treat or prevent disease
Corrects faulty genes causing disease
Can add, silence, or replace a gene
Targets somatic (body) or germline (inherited) cells
Emerging as a revolutionary medical treatment option

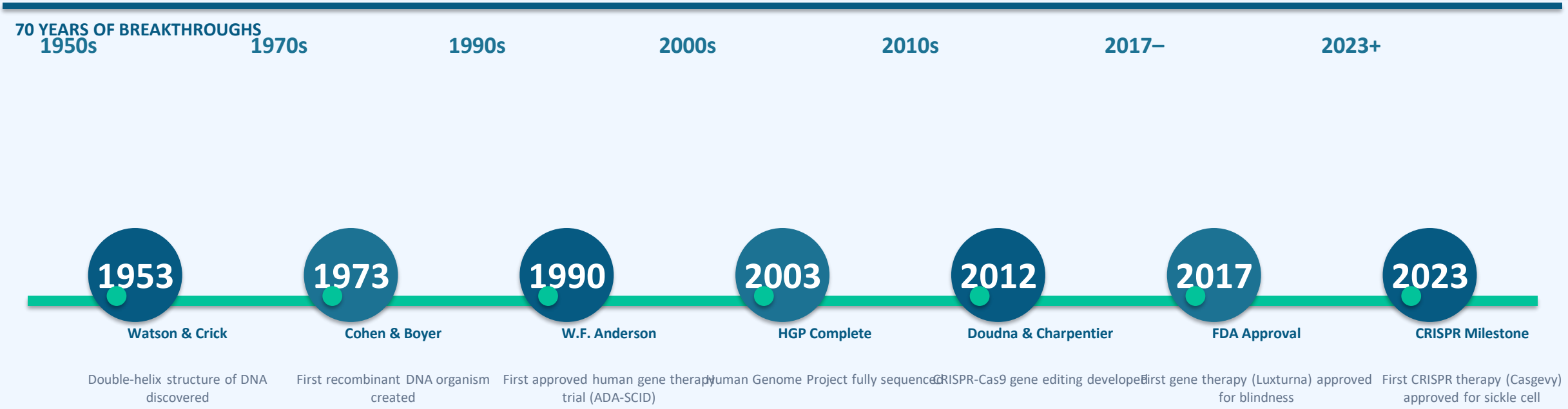
Genetic Engineering

Direct manipulation of an organism's DNA using biotechnology
Alters genes to produce desired traits
Includes recombinant DNA technology, CRISPR, cloning
Applied in medicine, agriculture, and research
Foundation of modern molecular biology

Key Relationship

Genetic Engineering is the broad toolbox of techniques; Gene Therapy is its targeted medical application — the science of rewriting DNA errors that cause disease.

History & Development: Key Milestones



Types of Gene Therapy

1 SOMATIC GENE THERAPY

Targets non-reproductive body cells

Target cells	Muscle, liver, lung, blood cells
Inheritance	Changes NOT passed to offspring
Safety profile	Lower ethical concern
Examples	CAR-T cell therapy, ADA-SCID treatment
Approved therapies	Zolgensma, Luxturna, Hemgenix
Delivery	In vivo (in the body) or ex vivo (outside body)
Current status	Clinically approved & widely researched

2 GERMLINE GENE THERAPY

Targets reproductive cells (eggs/sperm/embryos)

Target cells	Eggs, sperm, embryonic cells
Inheritance	Changes PASSED to all future generations
Safety concern	Extremely high — irreversible
Examples	CRISPR on human embryos (He Jiankui case)
Legal status	BANNED in most countries
Ethical issue	'Designer babies' risk and eugenics concern
Research use	Limited to lab models only (not clinical)

Methods of Gene Delivery

VIRAL VECTORS (Most Common – ~70% of trials)

Adeno-Associated Virus (AAV)	Retrovirus / Lentivirus	Adenovirus	Herpes Simplex Virus (HSV)
▶ Most widely used vector ▶ Low immune response ▶ Treats eye/muscle/nervous disorders ▶ Example: Zolgensma (SMA)	▶ Integrates into host DNA permanently ▶ Treats blood & immune disorders ▶ Risk: insertional mutagenesis ▶ Example: CAR-T cell therapy	▶ Efficient gene delivery, broad tropism ▶ Does NOT integrate into host DNA ▶ High immunogenicity drawback ▶ Used in COVID-19 vaccines	▶ Large gene cargo capacity ▶ Targets neurons effectively ▶ Used in pain & neurological research ▶ Less used clinically

NON-VIRAL METHODS (Emerging – Safer, Less Efficient)

Lipid Nanoparticles	Electroporation	Gene Gun	Naked DNA Injection
Coat DNA/mRNA in lipid bubble; used in mRNA COVID vaccines; efficient delivery	Electric pulse opens cell pores temporarily; allows DNA entry; used ex vivo	Coats gold beads with DNA, fired into tissue; used in agriculture & vaccines	Direct injection of plasmid DNA; simple but low efficiency; used in muscle

Genetic Engineering Techniques

01

Recombinant DNA Technology

- ▶ Cut DNA with restriction enzymes
- ▶ Insert gene into plasmid vector
- ▶ Transform into host cell (bacteria/yeast)
- ▶ Express desired protein at scale
- ▶ Example: Insulin production via E. coli

02

CRISPR-Cas9 Gene Editing

- ▶ Guide RNA directs Cas9 to target DNA
- ▶ Cas9 cuts double-stranded DNA precisely
- ▶ Cell repair inserts/deletes/replaces gene
- ▶ Cheap, fast, highly precise & programmable
- ▶ Nobel Prize 2020 – Doudna & Charpentier

03

Gene Cloning

- ▶ Isolate gene of interest from organism
- ▶ Insert into cloning vector (plasmid)
- ▶ Replicate in bacteria to produce many copies
- ▶ Used to study gene function & produce proteins
- ▶ Basis of PCR & molecular diagnostics

04

Zinc Finger & TALEN Editing

- ▶ Earlier protein-based gene editing tools
- ▶ Zinc fingers bind specific DNA sequences
- ▶ TALENs use TALE proteins + FokI nuclease
- ▶ More complex & expensive than CRISPR
- ▶ Still used in some therapeutic contexts

CRISPR-Cas9: Mechanism Explained

STEP 1

1

Design sgRNA

A 20-nucleotide guide RNA (sgRNA) is designed to match the target DNA sequence

STEP 2

2

sgRNA Binds Cas9

The sgRNA binds to Cas9 protein, forming an active ribonucleoprotein complex

STEP 3

3

DNA Scanning

The complex scans along the DNA strand, searching for the complementary sequence

STEP 4

4

Double-Strand Cut

Cas9 creates a precise double-strand break (DSB) at the target location

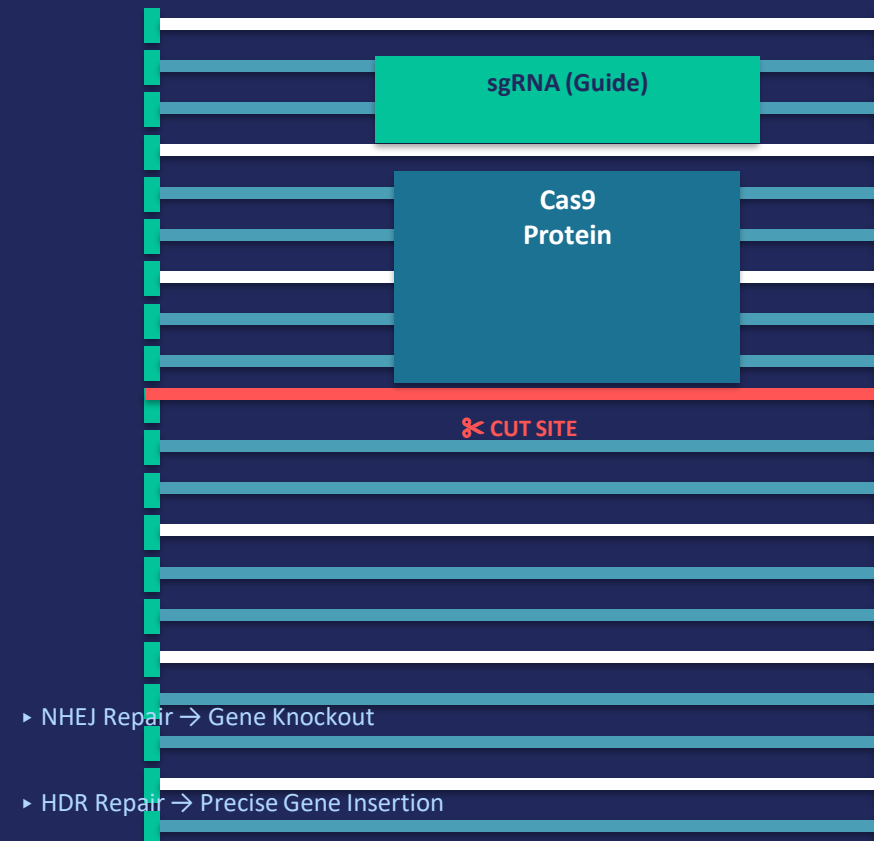
STEP 5

5

DNA Repair

Cell repairs via NHEJ (mutation) or HDR (precise gene insertion using a template)

CRISPR-Cas9 MECHANISM



► Nobel Prize 2020: Doudna & Charpentier

Medical Applications of Gene Therapy

Inherited Blood Disorders

Sickle cell disease & Beta-thalassemia: Casgevy (CRISPR-based) FDA-approved 2023. Hemophilia A/B treated with AAV vectors (Hemgenix).

- ▶ Casgevy – CRISPR sickle cell cure
- ▶ Zolgensma – SMA gene therapy
- ▶ Hemgenix – Hemophilia B (AAV5)

Neurological Conditions

Spinal Muscular Atrophy (SMA), Parkinson's, and Huntington's disease are active gene therapy targets.

- ▶ SMA treated with Zolgensma (one-time dose)
- ▶ Clinical trials for Parkinson's underway
- ▶ Huntington's: RNA interference trials

Eye Diseases

Luxturna – First FDA-approved gene therapy for inherited retinal dystrophy caused by RPE65 mutations. Restores functional vision.

- ▶ Luxturna (voretigene) – RPE65 mutation
- ▶ Age-related macular degeneration trials
- ▶ Stargardt disease research active

Cancer Immunotherapy

CAR-T cell therapy engineers patient's own T-cells to target and destroy cancer cells. Approved for leukemia & lymphoma.

- ▶ CAR-T: Kymriah & Yescarta approved
- ▶ Oncolytic virus therapy (T-VEC)
- ▶ Checkpoint inhibitor gene-based combos

Infectious Diseases

HIV: gene editing of CCR5 receptor confers resistance. COVID-19 mRNA vaccines use lipid nanoparticle gene delivery.

- ▶ mRNA vaccines (COVID-19, RSV, flu)
- ▶ HIV: CCR5-delta32 gene editing trials
- ▶ Hepatitis B/C gene silencing (RNAi)

Rare Genetic Syndromes

Hundreds of rare monogenic diseases (ADA-SCID, Gaucher, Pompe disease) are addressable with gene therapy.

- ▶ ADA-SCID: first ever gene therapy 1990
- ▶ Gaucher disease enzyme replacement
- ▶ Pompe disease AAV trials active

Genetic Engineering in Agriculture

~190M

Hectares of GM crops globally (2023)

29

Countries growing GM crops

\$20B+

Annual economic benefit to farmers

70%+

US soybean crop is GM

Pest & Disease Resistance

- ▶ Bt crops (Bacillus thuringiensis toxin gene)
- ▶ Kills insects without chemical pesticides
- ▶ Bt cotton, Bt corn, Bt brinjal (eggplant)
- ▶ Reduces crop losses up to 60%

Herbicide Tolerance

- ▶ Roundup Ready crops (glyphosate resistant)
- ▶ Farmers spray herbicide, only weeds die
- ▶ Soybean, maize, canola, cotton
- ▶ Controversial due to herbicide overuse

Improved Nutritional Value

- ▶ Golden Rice: beta-carotene (Vitamin A) added
- ▶ High-lysine corn for better protein profile
- ▶ Omega-3 enriched soybeans (SDA soybean)
- ▶ Addresses malnutrition in developing countries

Drought & Stress Tolerance

- ▶ DroughtGard maize survives water scarcity
- ▶ Salt-tolerant crops for marginal land use
- ▶ Cold-tolerant plants for early planting
- ▶ Critical for climate change adaptation

Animals & Aquaculture

- ▶ AquAdvantage salmon: faster growth rate
- ▶ Disease-resistant livestock via gene editing
- ▶ CRISPR hornless cattle (no dehorning needed)
- ▶ Pigs modified for organ xenotransplantation

Biofuel & Industrial Use

- ▶ High-starch/sugar crops for ethanol production
- ▶ Jatropha engineered for biodiesel yield
- ▶ Algae engineered to produce oil efficiently
- ▶ Reduce dependence on fossil fuels

Advantages & Benefits



Targeted Treatment

Corrects the root genetic cause of disease — not just symptoms. One-time treatment can be curative for life.



Reduced Drug Dependence

Patients may no longer need lifelong medication. Example: hemophilia patients avoiding factor replacement therapy.



Personalized Medicine

Therapy tailored to an individual's unique genome. Pharmacogenomics optimizes drug selection and dose.



Vaccine Revolution

mRNA vaccines (COVID-19) deliver genetic instructions for immunity — faster production, easier modification.



Food Security

GM crops resist pests, drought, and disease — boosting yields and reducing famine in vulnerable regions.



Disease Prevention

Germline editing could theoretically eliminate hereditary disorders (e.g., BRCA mutations) before birth.



Rapid Drug Production

Recombinant proteins (insulin, erythropoietin, growth hormone) produced cheaply and abundantly in bacteria.



Research Acceleration

CRISPR enables rapid gene function studies. Animal models of disease created in weeks instead of years.

Risks, Safety Concerns & Ethical Debates

SAFETY RISKS

► Immune Reactions

Viral vectors may trigger severe immune responses (e.g., Jesse Gelsinger death, 1999)

► Off-Target Effects

CRISPR may cut unintended genomic sites, causing mutations or cancer

► Insertional Mutagenesis

Retroviral integration near oncogenes can cause leukemia (observed in early ADA-SCID trials)

► Long-term Unknowns

Long-term safety of gene modifications not yet fully known

► Delivery Challenges

Reaching all target cells efficiently, especially in large organs, remains difficult

ETHICAL DEBATES

► Designer Babies

Selecting traits (intelligence, appearance) raises eugenics concerns and inequality

► Informed Consent

Future generations affected by germline edits cannot consent to changes

► Access & Equity

Gene therapies cost \$1–3 million; only wealthy nations/patients can afford them

► Playing God

Philosophical/religious debates about humans altering the fundamental code of life

► Dual-Use Risk

CRISPR knowledge could be misused for bioweapons or harmful pathogens

REGULATORY OVERVIEW

FDA (USA)

Strict IND & BLA approval needed. Monitors all gene therapy trials.

EMA (Europe)

Centralized ATMP (Advanced Therapy) approval process.

WHO Guidelines

Global ethical framework; opposes germline editing in humans.

Moratorium

International consensus: no clinical germline editing until safe & ethical.

He Jiankui

Chinese researcher jailed 3 years for illegal CRISPR baby experiments (2018).

Current Research & Future Scope

AI & Machine Learning in Genomics

- ▶ AlphaFold2 predicted protein structures of all known proteins
- ▶ AI designs optimized guide RNAs for CRISPR
- ▶ ML predicts off-target CRISPR cuts before editing
- ▶ AI-driven drug target identification from genomic data

Next-Gen Gene Editing Tools

- ▶ Base editing: changes single DNA letters without cuts
- ▶ Prime editing: 'search and replace' for any mutation
- ▶ Epigenome editing: modifies gene expression without changing DNA
- ▶ CRISPR-Cas12 & Cas13: DNA and RNA targeting

In Vivo Gene Editing

- ▶ Delivering CRISPR directly into living patients via LNPs
- ▶ NTLA-2001: CRISPR therapy for transthyretin amyloidosis
- ▶ Intellia & Regeneron in vivo liver-targeted trials
- ▶ Goal: treat diseases without removing cells from body

Whole Genome Sequencing (WGS)

- ▶ Cost dropped from \$3 billion (2003) to <\$200 today
- ▶ Newborn genomic screening programs being piloted
- ▶ Prenatal genetic diagnosis at single-cell level
- ▶ Population genomics for disease risk profiling

Global Gene Therapy Access

- ▶ mRNA platform allows rapid vaccine & therapy development
- ▶ Gates Foundation funds gene therapy for malaria & HIV
- ▶ Low-cost CRISPR kits now used in developing countries
- ▶ WHO 2030 roadmap for equitable genomic medicine access

Predictions for 2030–2040

- ▶ >50 gene therapies routinely approved worldwide
- ▶ CRISPR used to cure HIV, Alzheimer's, common cancers
- ▶ Personalized cancer vaccines based on tumor genome
- ▶ Potential extension of healthy lifespan via longevity genes

Real-World Case Studies & Success Stories

CASE 1

Ashanthi DeSilva – (ADA-SCID, 1990)

DISEASE

Adenosine Deaminase Deficiency (SCID) – severe combined immunodeficiency, no immune system

TREATMENT

First approved human gene therapy. Retroviral vector delivered functional ADA gene into T-cells ex vivo

OUTCOME

Immune system restored. Ashanthi is alive and healthy 30+ years later — the first gene therapy success story

CASE 2

Sickle Cell Disease – (Casgevy, 2023)

DISEASE

Sickle Cell Disease – mutant HBB gene causes deformed red blood cells, severe pain crises

TREATMENT

CRISPR-Cas9 edits BCL11A gene to reactivate fetal hemoglobin (HbF), compensating for defective adult hemoglobin

OUTCOME

FDA-approved Nov 2023. 94% of patients pain-crisis-free at 1 year. First approved CRISPR therapy globally

CASE 3

Inherited Blindness – (Luxturna, 2017)

DISEASE

Leber Congenital Amaurosis – RPE65 gene mutation causes progressive blindness from childhood

TREATMENT

AAV2 vector delivers functional RPE65 gene directly into subretinal space via surgical injection

OUTCOME

FDA-approved 2017. Restored functional vision in both eyes. Patient can navigate independently in dim light

CASE 4

Spinal Muscular – Atrophy (Zolgensma)

DISEASE

SMA Type 1 – SMN1 gene deletion causes motor neuron death; children unable to breathe/swallow by age 2

TREATMENT

AAV9 vector delivers SMN1 gene via single IV infusion (onasemnogene abeparvovec)

OUTCOME

Most expensive drug ever at \$2.1M — but one-time cure. 91% of treated infants alive at age 2 (vs <8% untreated)

Key Takeaways



Gene therapy corrects disease at the genetic root, offering potential one-time cures for hereditary conditions



CRISPR-Cas9 has revolutionized genetic engineering — precise, affordable, and programmable gene editing



Approved therapies (Casgevy, Zolgensma, Luxturna, CAR-T) demonstrate the real-world power of gene therapy



Genetic engineering in agriculture improves food security through GM crops with enhanced traits



Ethical oversight is essential — germline editing, designer babies, and access inequality demand global governance



The integration of AI, advanced vectors, and base/prime editing promises an extraordinary future for medicine

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